Reviewer Pack — Grothendieck-Riemann-Roch Theorem and Resolution Tree Width …

Entry fba35e5a136c · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Grothendieck-Riemann-Roch Theorem and Resolution Tree Width for Tseitin Formulas

Entry ID: `fba35e5a136c` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-05-21 13:10:16 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Geometry: Grothendieck-Riemann-Roch Theorem **Field B** (complexity object): Complexity Theory: Resolution Tree Width of Tseitin Formulas

Statement:

> ['For any given Tseitin formula, the resolution tree width is at least exponential in the number of variables.', 'More specifically, for a Tseitin formula with m variables and n clauses, the resolution tree width is Ω(2^m).', 'This implies that Tseitin formulas are hard to solve using resolution refutations.']

Rationale (proposer's reasoning):

> ['The Grothendieck-Riemann-Roch Theorem provides a relationship between topological invariants of algebraic varieties and their cohomology. By applying this theorem to the algebraic variety associated with the Tseitin formula, we can potentially derive lower bounds on the resolution tree width.', 'This connection could expose new structural insights into the complexity of solving Tseitin formulas using resolution refutations.', 'The Riemann-Roch Theorem has previously been used in other contexts to obtain circuit complexity lower bounds.']

Taxonomy category: `TSEITIN_RES` (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `d231a45386fd892d`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> For a Tseitin formula with m variables, if the measured resolution tree width is at least 2^m - 1 for all 30 seeds, this supports the conjecture.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM_1	LLM_2
RELATIVIZATION	UNCERTAIN	1.00	HITS	SAFE
NATURAL_PROOFS	SAFE	0.90	SAFE	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
KARP_LIPTON	SAFE	1.00	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=1, elapsed=0.1s

5.1 Generated Python source

```python
import random
import math
from fractions import Fraction
import sys

def generate_tseitin_formula(m, n):
    variables = [f'x{i}' for i in range(m)]
    clauses = []

    # Generate clauses for each variable
    for i in range(m):
        clause = [variables[i]]
        for j in range(n):
            if random.choice([True, False]):
                clause.append(f'~{variables[j]}')
        clauses.append(clause)

    # Generate binary clauses between variables
    for i in range(m):
        for j in range(i + 1, m):
            clauses.append([f'~{variables[i]}', f'{variables[j]}'])
            clauses.append([f'~{variables[j]}', f'{variables[i]}'])

    return clauses

def resolution_tree_width(clauses):
    variables = set()
    for clause in clauses:
        for literal in clause:
            if literal.startswith('~'):
                variables.add(literal[1:])
            else:
                variables.add(literal)
```

```
    variables = sorted(variables)
    variable_map = {v: i for i, v in enumerate(variables)}

    tree = [[] for _ in range(len(variables))]

    def add_clause(clause):
        resolvents = []
        for literal in clause:
            if literal.startswith('~'):
                negated_var = literal[1:]
                if negated_var in variable_map:
                    resolvent = (negated_var, literal)
                    resolvents.append(resolvent)

        for i in range(len(resolvents)):
            for j in range(i + 1, len(resolvents)):
                var1, lit1 = resolvents[i]
                var2, lit2 = resolvents[j]
                if var1 != var2:
                    new_clause = [lit1[1:], lit2[1:]]
                    add_clause(new_clause)

        for var, literal in resolvents:
            tree[variable_map[var]].append(variable_map[literal])

    for clause in clauses:
        add_clause(clause)

    max_width = 0
    visited = [False] * len(tree)

    def dfs(node):
        if visited[node]:
            return 0
        visited[node] = True
        width = 1
        for neighbor in tree[node]:
            width = max(width, dfs(neighbor) + 1)
        return width

    for i in range(len(variables)):
        max_width = max(max_width, dfs(i))

    return max_width

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n = random.choice([5, 10, 15, 20, 30, 40])
    m = n // 2
    clauses = generate_tseitin_formula(m, n)

    width = resolution_tree_width(clauses)

    return {
        "metric_name": "resolution_tree_width",
```

```
        "metric_value": width,
        "instances_tested": len(clauses),
        "conjecture_holds": width >= 2**m - 1,
        "counterexample": "" if width >= 2**m - 1 else f"Formula with m={m}, n={n} failed"
    }

if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] or list(range(2, 30*37, 127))[:30]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_width = sum(res["metric_value"] for res in results) / len(results)
    std_width = math.sqrt(sum((res["metric_value"] - mean_width)**2 for res in results) / len(resu
    support_fraction = sum(1 for res in results if res["conjecture_holds"]) / len(results)

    if all(res["conjecture_holds"] for res in results):
        print(f"RESULT: SUPPORTED mean={mean_width} std={std_width} support_fraction={support_frac
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_width} std={std_width} support_fraction={support_frac
    else:
        first_failing_seed = next((res["seed"] for res in results if not res["conjecture_holds"]),
        print(f"RESULT: FALSIFIED counterexample=\"Formula with m={m}, n={n} failed\" first_failin
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5fe8051f.py", line 113, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5fe8051f.py", line 96, in run_trial
    clauses = generate_tseitin_formula(m, n)
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5fe8051f.py", line 27, in generate_tsei
    clause.append(f'~{variables[j]}')
                      ~~~~~~~~~^^^
IndexError: list index out of range
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: `INCONCLUSIVE`

Reasoning:

> The test code crashed before producing data, which means it did not complete its execution to measure the resolution tree width. | next: Re-run the test with proper error handling and ensure that it completes without crashing.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	15493
2	preregistration	ollama_remote	glm4:latest	0	0	9381
3	novelty	ollama_remote	glm4:latest	0	0	8204
4	novelty	ollama_remote	glm4:latest	0	0	8759
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11012
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7733
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10622
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11708
9	judge	ollama_remote	glm4:latest	0	0	11407

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 94320 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/fba35e5a136c.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/fba35e5a136c.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/fba35e5a136c.tar.gz` (if generated)